
A Hybrid Physics-ML Model for Atmospheric Column Forecasting: A mini-NeuralGCM

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Abstract

This project implements a simplified one-dimensional atmospheric column model that captures the core innovation of NeuralGCM: integrating machine learning to learn subgrid parameterizations while enforcing explicit physical constraints. The model predicts temporal evolution of temperature and humidity by combining physics-based numerical methods with neural network parameterizations of convection and surface fluxes.

1 Overview

Recent hybrid weather prediction models like NeuralGCM replace computationally expensive physical parameterizations with neural networks while preserving conservation laws. This project implements a simplified 1D atmospheric column model demonstrating this approach: physics-based components (vertical diffusion, radiative heating) are combined with learned parameterizations (convection, surface fluxes) to predict temporal evolution of temperature and humidity profiles.

2 Dataset and Preprocessing

Vertical atmospheric profile data from St. John's, Newfoundland will be sourced from ERA5 reanalysis. Required variables are temperature and humidity profiles, surface pressure, and incoming solar radiation. Data will be interpolated to standard pressure levels and split chronologically into train/validation/test sets.

3 Model Architecture

The **physics module** explicitly simulates vertical diffusion (finite differences), radiative heating (Newtonian cooling), and hydrostatic balance. The **learned module** is a multilayer perceptron predicting convective heating rates and surface fluxes from current column state (T, q, p) with physically-constrained outputs (non-negative fluxes, bounded heating). These tendencies are combined as $\frac{\partial \mathbf{x}}{\partial t} = \mathcal{F}_{\text{physics}}(\mathbf{x}) + \mathcal{F}_{\text{learned}}(\mathbf{x}; \boldsymbol{\theta})$ and time-stepped using Runge-Kutta.

4 Training and Evaluation

Training minimizes $\mathcal{L} = \text{MSE}(\mathbf{x}_{\text{pred}}, \mathbf{x}_{\text{obs}}) + \lambda_E \mathcal{L}_{\text{energy}} + \lambda_P \mathcal{L}_{\text{physics}}$ where energy conservation and physical constraints (non-negative humidity) are enforced. Multi-step prediction prevents error accumulation. Four

baselines isolate contributions: persistence, physics-only, ML-only (LSTM), and hybrid (proposed). Evaluation uses RMSE/MAE for profiles, energy conservation error, physical constraint violations, and forecast skill degradation over multiple step lead times.

5 Expected Outcomes and Impact

A working hybrid model will demonstrate improved skill over pure-physics and pure-ML baselines, revealing where ML augments physical modeling and where constraints ensure stability. This provides hands-on experience with NeuralGCM's fundamental challenge which was integrating data-driven learning with physical constraints directly supporting thesis work on NeuralGCM applications to Canadian climate data.